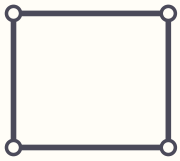
STEM Little Explorers

Which shapes hold, and which fold

Build each frame out of drinking straws or craft sticks. Push a joint sideways. A frame that keeps its shape is rigid. A frame that leans over into a new shape is not, and no amount of stronger material fixes it.

How to build a frame. Cut straws to length and join them with pipe cleaners bent into a U, or overlap craft sticks and push a split pin through both. The joints must be able to turn. That is the point: a rigid frame stays rigid even when its corners are free to pivot.

Square

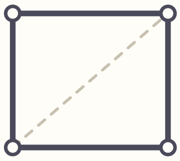


Joints: **4** Bars: **4** Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

Square with one diagonal

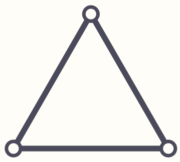


Joints: **4** Bars: **5** Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

Triangle

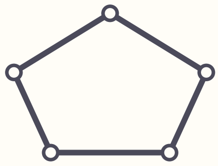


Joints: **3** Bars: **3** Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

Pentagon

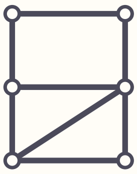


Joints: **5** Bars: **5** Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

Two storeys, one diagonal



Joints: **6** Bars: **8** Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

Draw a frame of your own

Joints: ____ Bars: ____ Needs at least: ____

I predict: ☐ holds ☐ folds

What happened: ____________________

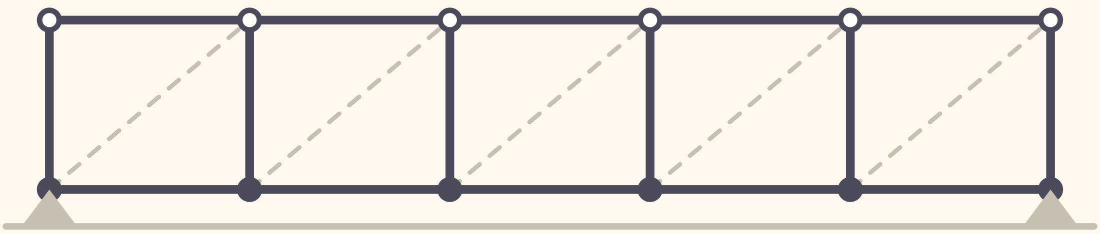
The counting rule. A flat frame with J joints needs at least 2 × J − 3 bars before it can be rigid. Fewer than that and it is certainly floppy. Enough bars is not a promise on its own, because they still have to be in the right places, but it tells you where to look. Fill in the third column and compare it with what your hands found.

Build a bridge that holds a load

Draw a diagonal into each square below, then build the same design and test it. Add coins one at a time until the deck sags, and write down how many it took.

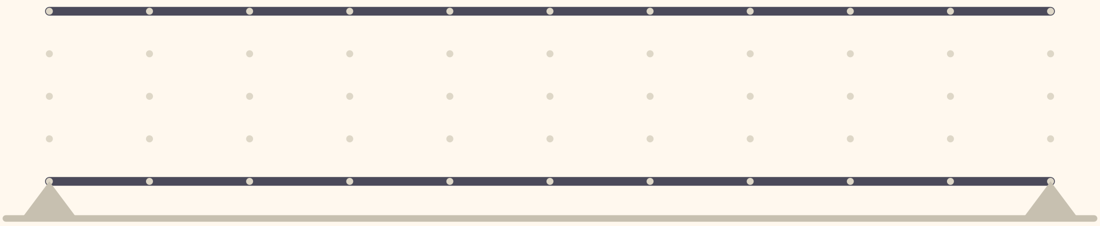
1. Plan the bracing

Each dashed line is a diagonal you could add. Trace the ones you want. Which way each diagonal leans does not change whether the bridge keeps its shape.



2. Your own span

Same two chords, no bays marked. Draw your own bars, then count the joints and check them against the rule.



3. Load log

DESIGN	DIAGONALS USED	COINS HELD	WHERE IT FAILED
No diagonals			
Every other bay			
Every bay			
My own design			

Think about it. The bridge almost never breaks a bar. It gives way at a joint, or a square leans over and takes the deck with it. Watch closely and write down where yours went first.